Tap and Die Set



Category: **Hand Tools - Cutting & Shaping** Type: **Short SOP** Priority: **Medium** 11 labs

Covers these lab items: Die set complete; Die wrench; Tap & Die Set; Tap or die; Tap Wrench

What it is used for

Use a tap and die set to cut or repair internal threads in holes and external threads on rods or bolts.

How to use

1. Select the correct tap or die for the thread size, pitch, and material.
2. Inspect the cutting edges and handles for damage or wear.
3. Secure the workpiece firmly and prepare the hole or rod to the required size.
4. Apply suitable cutting fluid when appropriate for the material.
5. Align the tap or die squarely with the workpiece before turning.
6. Turn the tool gradually, reversing occasionally as required to clear chips.
7. Remove the tool carefully and inspect the finished thread with a suitable fastener or gauge.
8. Clean the tools and remove metal chips after use.

⚠ Safety

- Wear safety glasses and keep hands clear of sharp cutting edges.
- Clamp the workpiece securely; do not hold it by hand while cutting threads.
- Use the correct tool size and avoid excessive turning force.
- Remove chips with a brush, not bare fingers.

Care & storage

- Clean and lightly protect cutting tools against rust after use.
- Store taps, dies, and handles in their labeled case.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.